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Deriving Proton Pitch Angle Distributions Using Sentinel 6 – MF NGRM Observations

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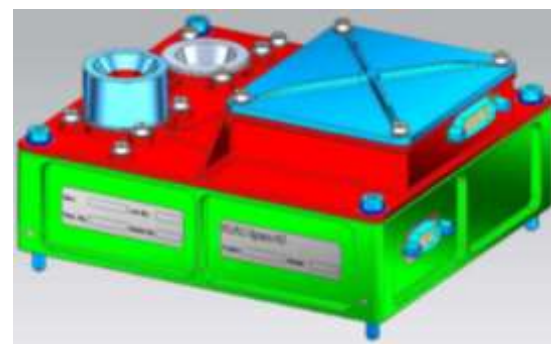
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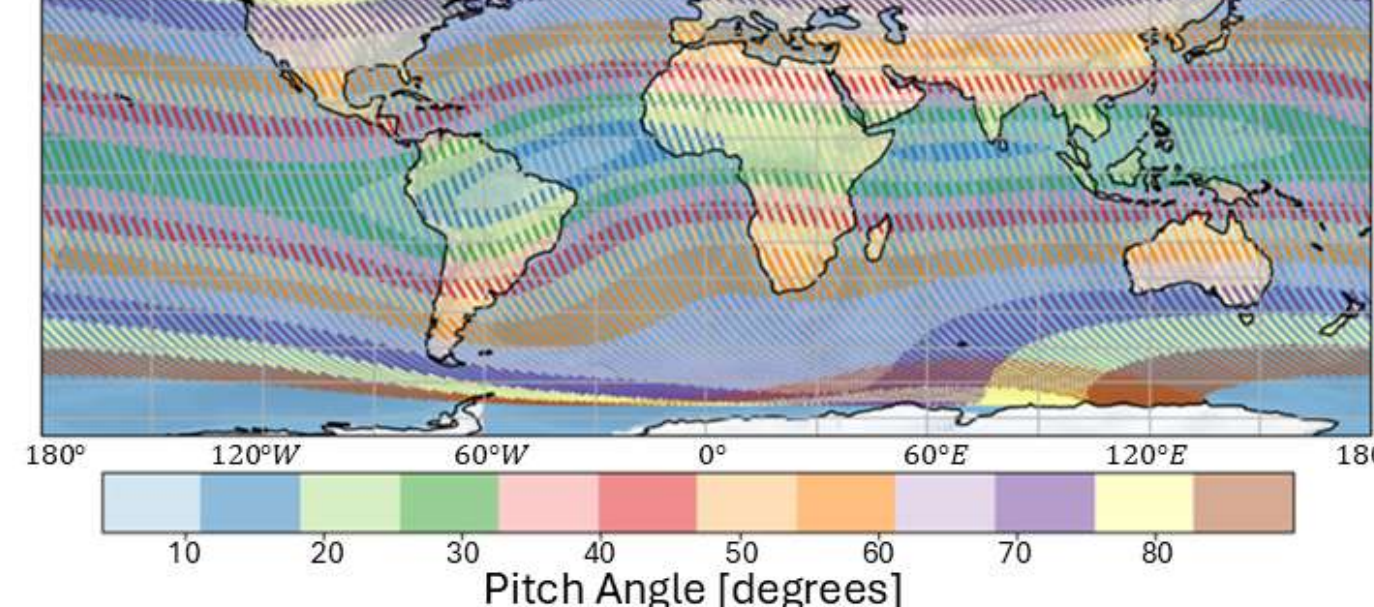
Motivation and Data

The inner radiation belt exhibits strongly anisotropic pitch angle distributions (PADs), which are a particularly powerful diagnostic for understanding the physical processes governing particle behaviour in the inner magnetosphere.

Sentinel 6 (S6) is an Earth observation mission in near-polar, LEO orbit, that carries a **Next Generation Radiation Monitor (NGRM)**, which measures electrons and protons across multiple energy ranges [1-2]. However, the detector is mounted in a fixed orientation, (the detector bore site is aligned along the spacecraft flight direction), which fixes the pitch angle of the instrument. The orbit has a 9 day repeat cycle, which provides the minimum temporal binning limit.

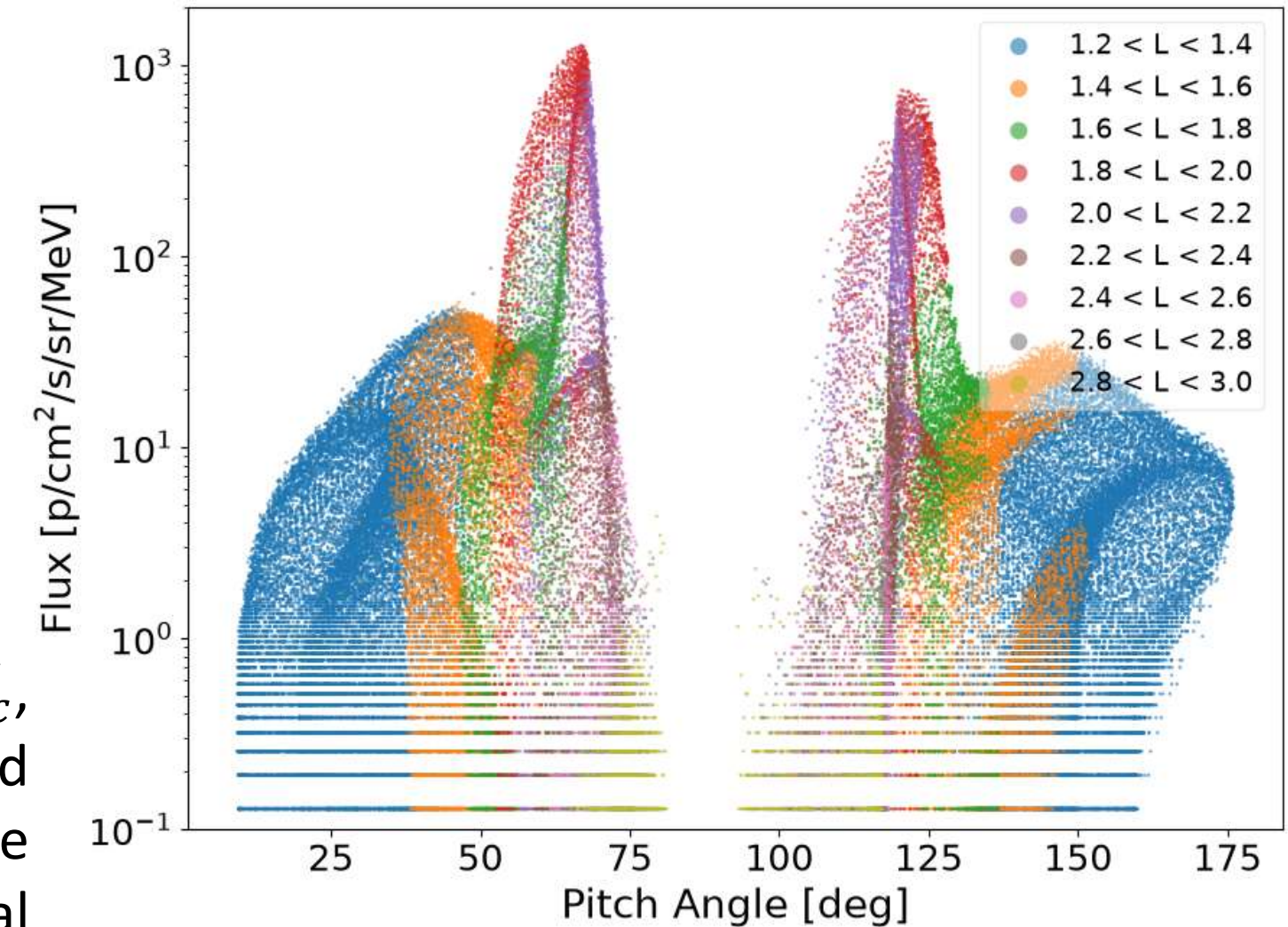


- We utilise the **Level 1 Version 1 Proton Differential Flux Dataset** (Produced by SPARC [3], publicly available through SWE HAPI [4]).
- We focus on **9 MeV protons** (SDSS-2).
- Most measurements cluster within two narrow pitch-angle ranges, approximately $10^\circ - 75^\circ$ and $100^\circ - 170^\circ$ (See Figure 3).
- Northbound and southbound passes sample different pitch-angle intervals (See Figure 4).



Figures 3: Example of the PAs sampled during (a) northbound and (b) southbound passages of a quiet month.

- Pitch angle is computed as $PA = \cos^{-1} \hat{B} \cdot \hat{X}_{sc}$, where $\hat{B}$ is the magnetic field vector calculated from the IGRF magnetic field model with the Olson and Pfitzer (1977) quiet time external model, and $\hat{X}_{sc}$ is the spacecraft +X.
- Key assumption:** northbound and southbound passes sample the same underlying population in the same geographic region over the subsampled periods.



Figures 4: PAD of the measured proton fluxes at 9 MeV, where colour indicates the different L-shells.

GRAS/GEANT4 Simulations and Fitting Process

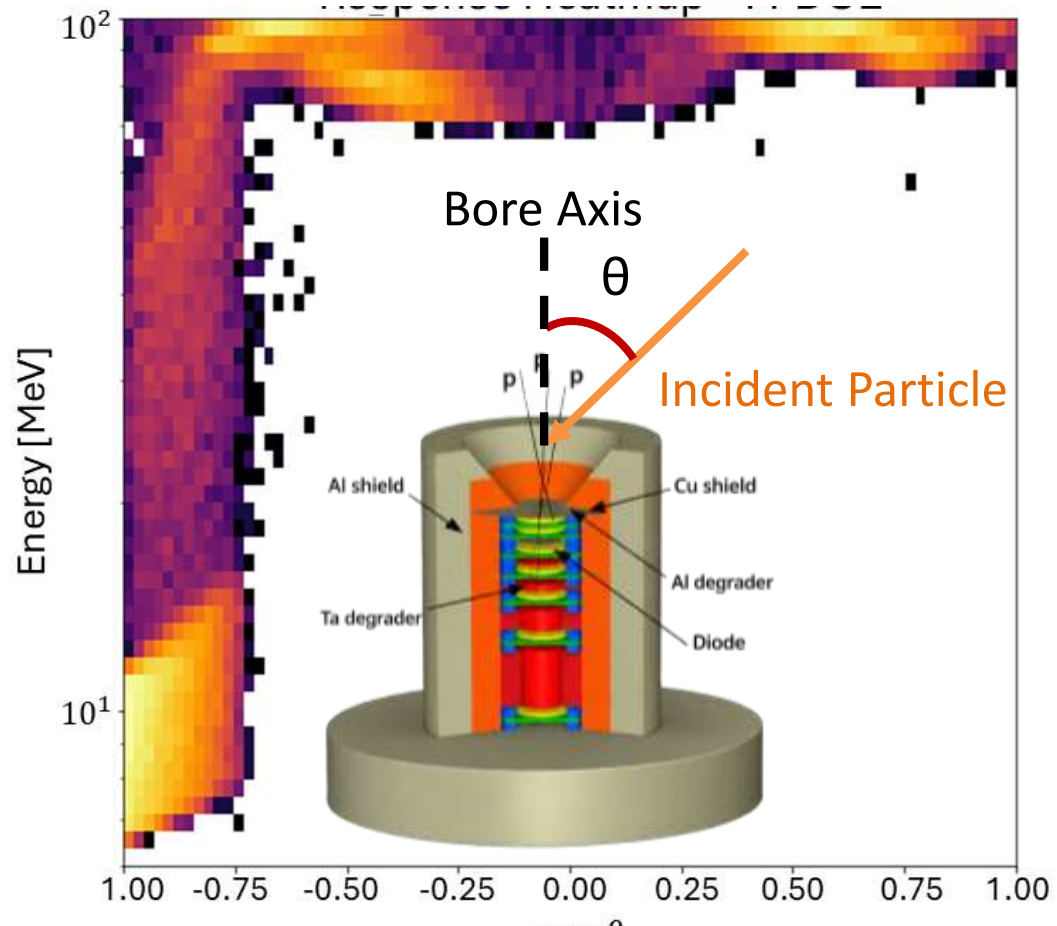


Figure 5: Heat map of the channel's response as a function of incident angle and proton energy + Schematic of how θ is measured

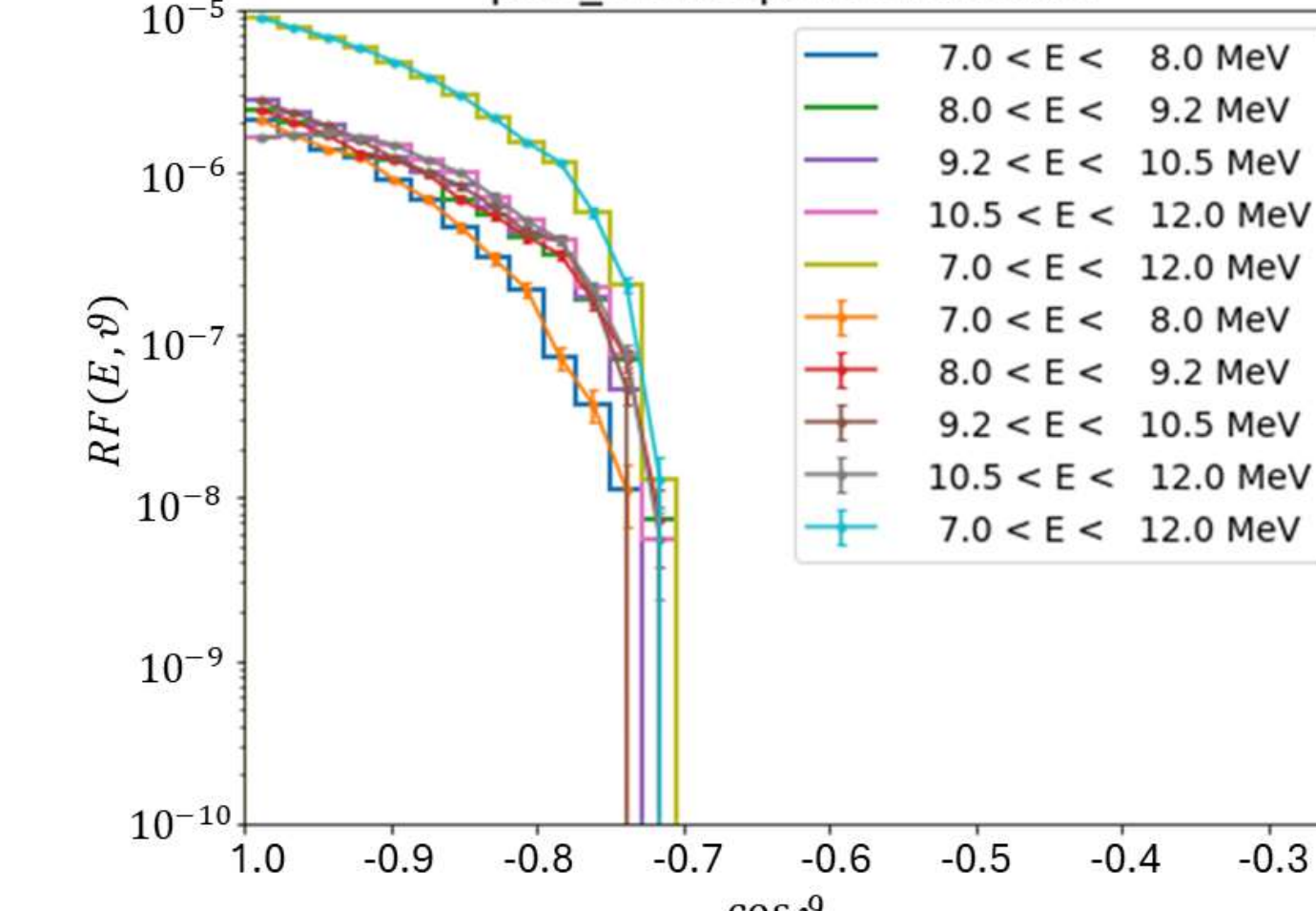


Figure 6: Histogram of the angular detector response as a function of $\cos \theta$. Colour corresponds to different energy bins.

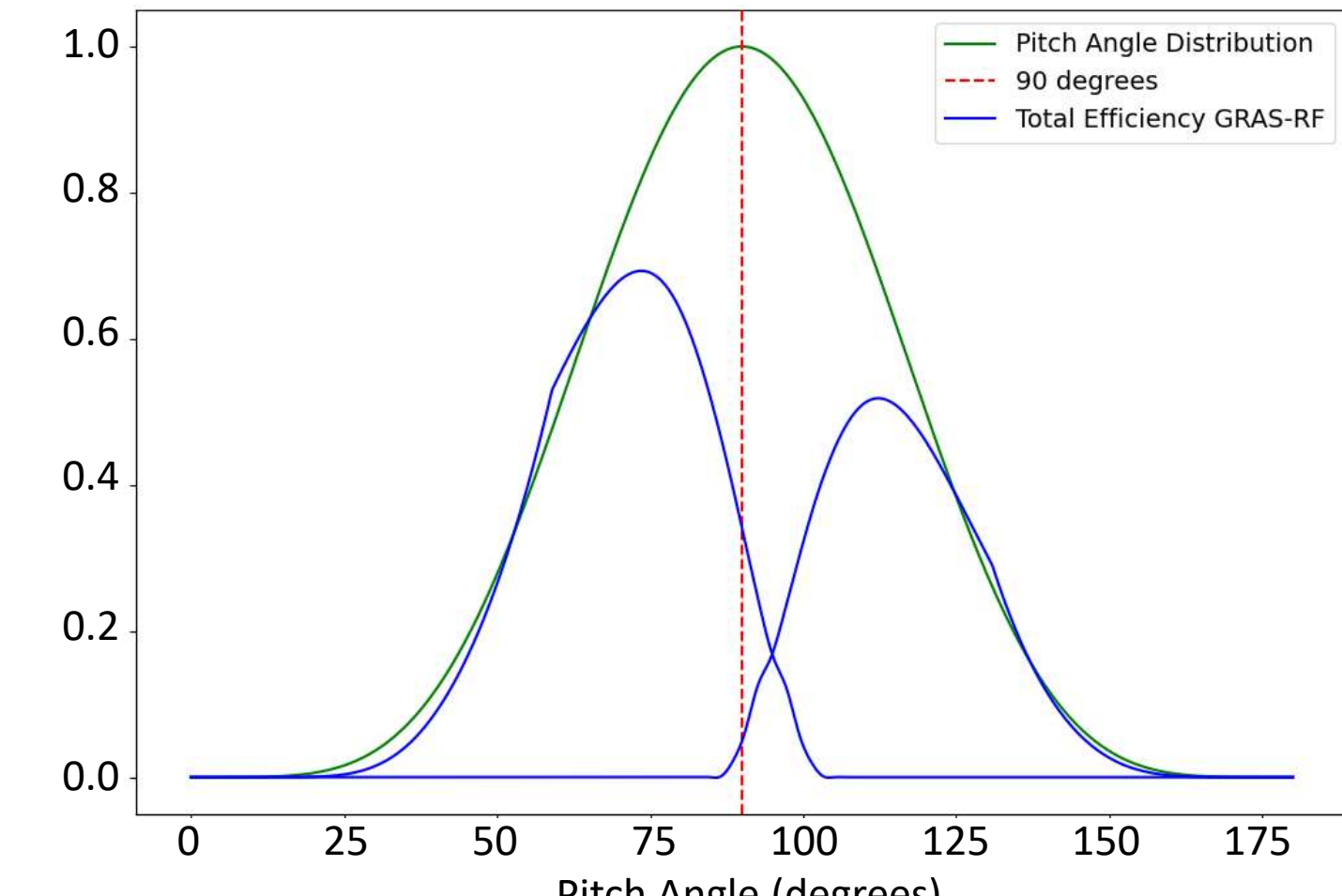


Figure 7: Green curve: the best-fit PAD, blue curve: computed total efficiency. Dashed red line indicates 90° .

- Developed high-fidelity angular response function for NGRM using **ESA's GEANT4 Radiation Analysis for Space (GRAS) toolkit** [5].
- Full GDML model of NGRM's proton/heavy-ion subsystem (SDSS/HEP): 7 silicon detectors separated by degraders, 40° aperture, side-shielded by Cu/Al.
- Physics: EM Standard Opt4 (electromagnetic) + QBBC (hadronic), 0.1mm cuts, 1-10,000 MeV energy range.
- 10^9 primaries emitted isotropically (cosine distribution, $0^\circ - 90^\circ$) from a 25cm sphere fully enclosing the detector.
- Analysis restricted to 7–12 MeV, where SDSS-2 sensitivity and statistics are strongest - limiting overestimation from side/back attenuation not captured in the free-flight.
- Data split into 9-day segments (Jan 2024 – Jan 2026) and $4^\circ \times 4^\circ$ lat–lon spatial bins; passes are separated into northbound ($PA > 90^\circ$) and southbound ($PA < 90^\circ$) samples.
- PAD modelled as $f(\alpha) \propto \sin^n \alpha$; detector efficiency for a given n: $E_{model}(N) = \frac{\int_0^{180^\circ} R(\alpha) \sin^N \alpha d\alpha}{\int_0^{180^\circ} \sin^N \alpha d\alpha}$
- For each bin: compute the observed north/south flux ratio and compare to the modelled efficiency ratio across a grid of n (1-12, step 0.25). **Best-fit n:** $R_{observed} \approx R_{modelled}(n)$.

Results

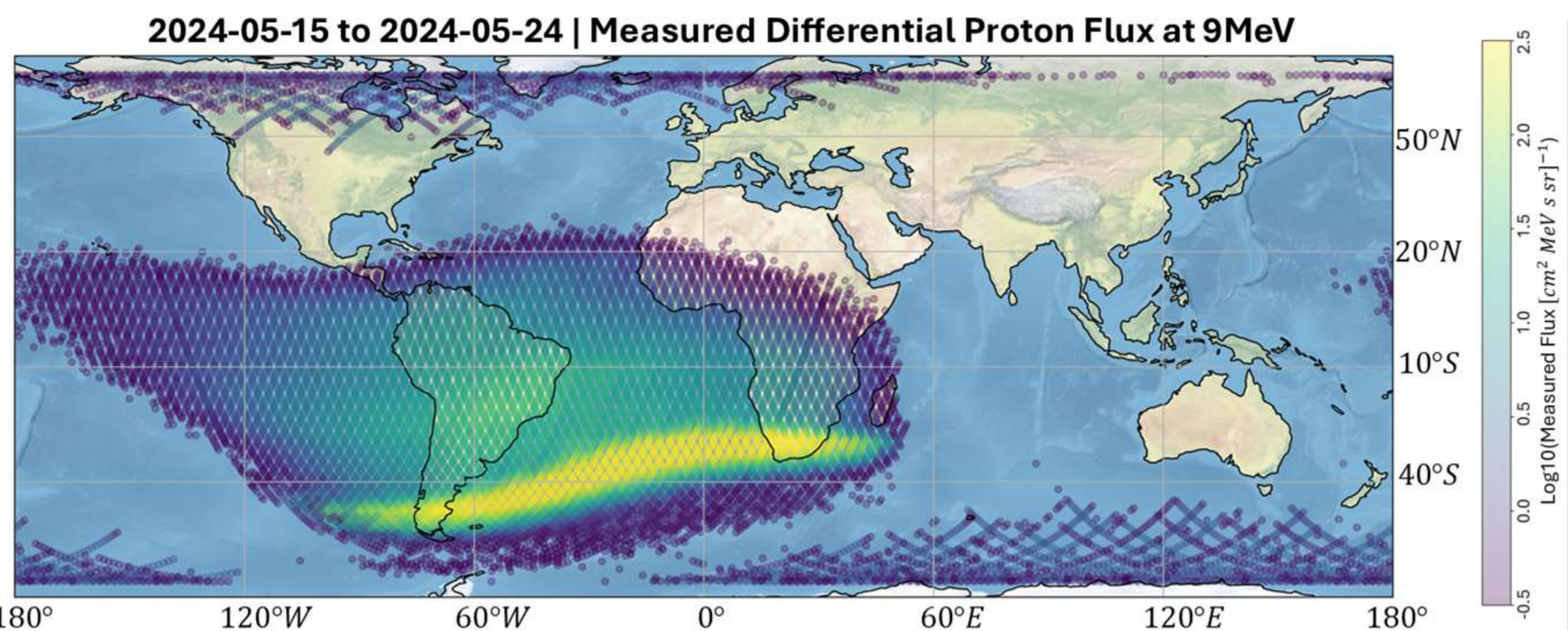


Figure 8: Flux-map with colour showing S6-MF/NGRM Level 1 Version 1 differential fluxes in log axis, for 15-24 March 2024. The enhancement of the SAA (corresponding to the outer edge of the inner radiation belt at low altitudes) is visible.

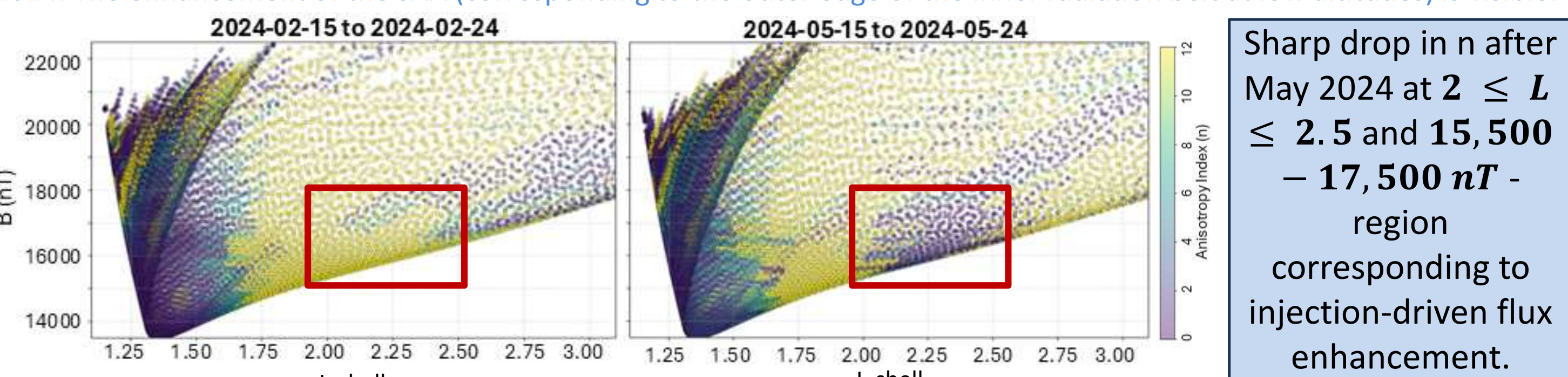


Figure 11: Fitted PAD exponent n in B-L space for the time periods (right) 15--24 February 2024, (left) 15--24 March 2024.

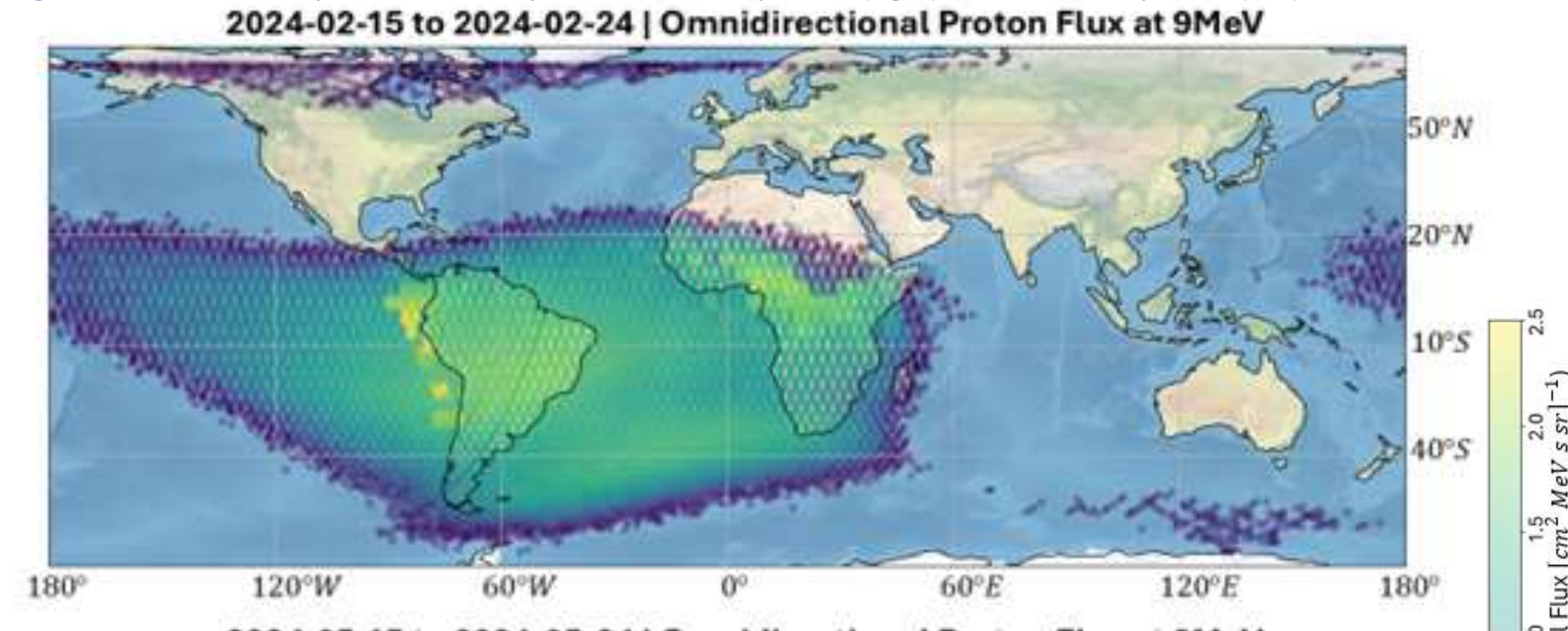


Figure 12: Derived omnidirectional proton flux at 9 MeV, for (top) 15-24 February 2024, (bottom) 15-24 March 2024.

- Flux enhancement of **at least an order of magnitude** following the May 2024 storm.
- Localized "hotspots" near the west coast of South America and central Africa are **fitting artifacts, not physical** - there, the detector look angle is nearly parallel to the field lines, therefore the fitting process leads to overestimations of the flux.

References:
 [1] Desorgher et al.(2013), "The Next Generation Radiation Monitor–NGRM", IEEE/NSS/MIC.
 [2] Sandberg et al.(2022), "First Results and Analysis From ESA Next Generation Radiation Monitor Unit Onboard EDRS-C", IEEE/TNS, 69(7)
 [3] Sandberg (2023), S6-MF/NGRM Level 1 Data Description, Tech. Rep., ESA.
 [4] ESA, Next Generation Radiation Monitor (NGRM), <https://swe.ssa.esa.int/en/ngrm>
 [5] Santin G. et al.(2005), "GRAS: a general-purpose 3-D Modular Simulation tool for space environment effects analysis", IEEE/TNS, 52(6).

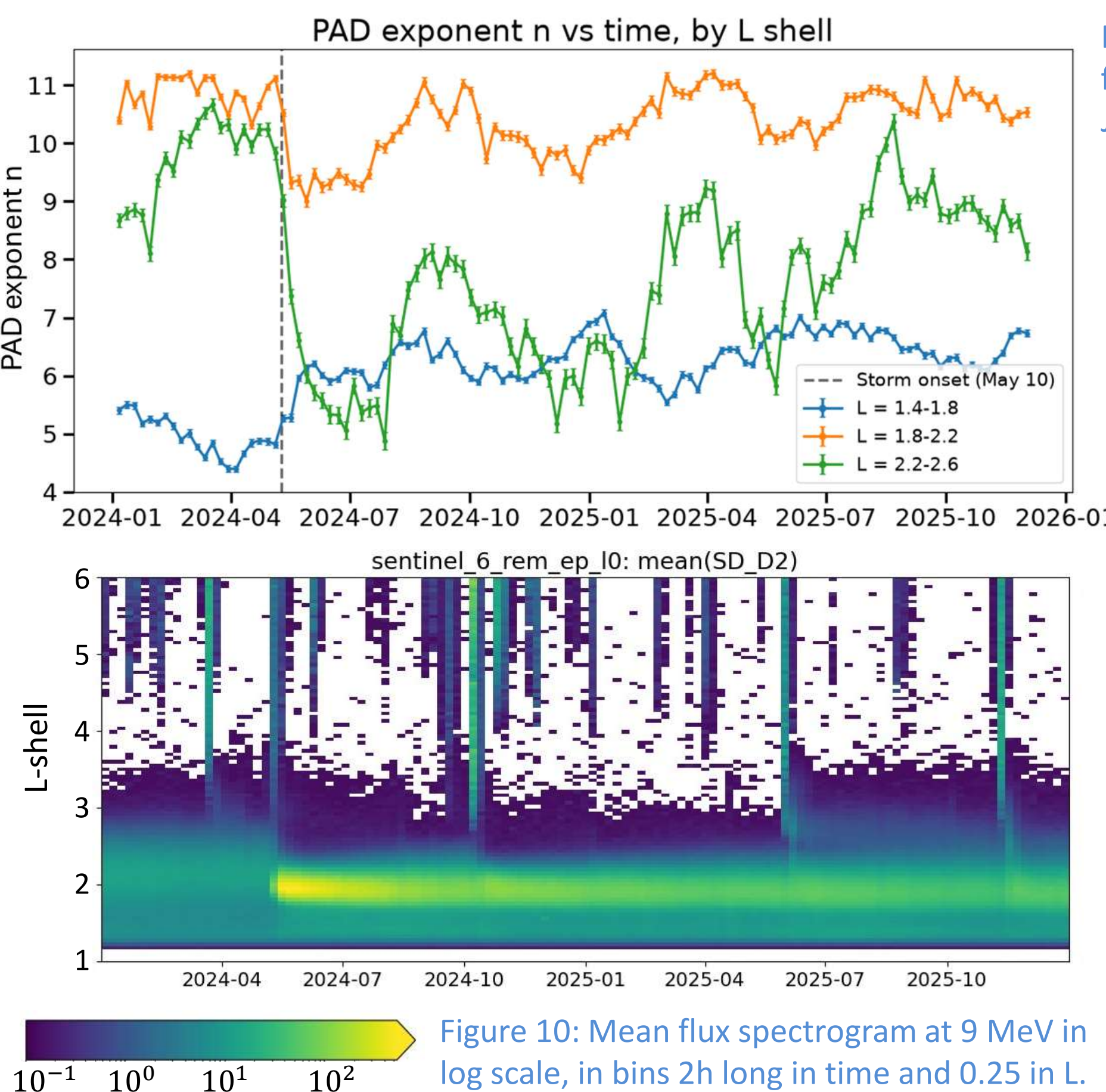


Figure 9: Fitted PAD exponent n as a function of time, with error bars, for January - August 2024, grouped by L-shell.

- Heart of inner belt** ($L = 1.4 - 1.8$): $n \approx 5.5 \pm 1$, close to isotropy and largely insensitive to geomagnetic activity.
- Outer edge of inner belt** ($L = 1.8 - 2.2$): $n \approx 11 \pm 0.5$ in quiet time, dropping by ~ 2 units during the May 2024 superstorm.
- Approaching slot region** ($L = 2.2 - 2.6$): drops from $n \approx 10$ to $n \approx 6$ during the storm.
- Post-storm recovery of n is **non-monotonic** at $L = 2.2 - 2.6$, repeatedly interrupted by subsequent injections.

$$\Phi_{\text{omni},X} = \frac{F_X}{E_X(n)}, \text{ for } X \in \{N, S\},$$

where n is the best-fit n.

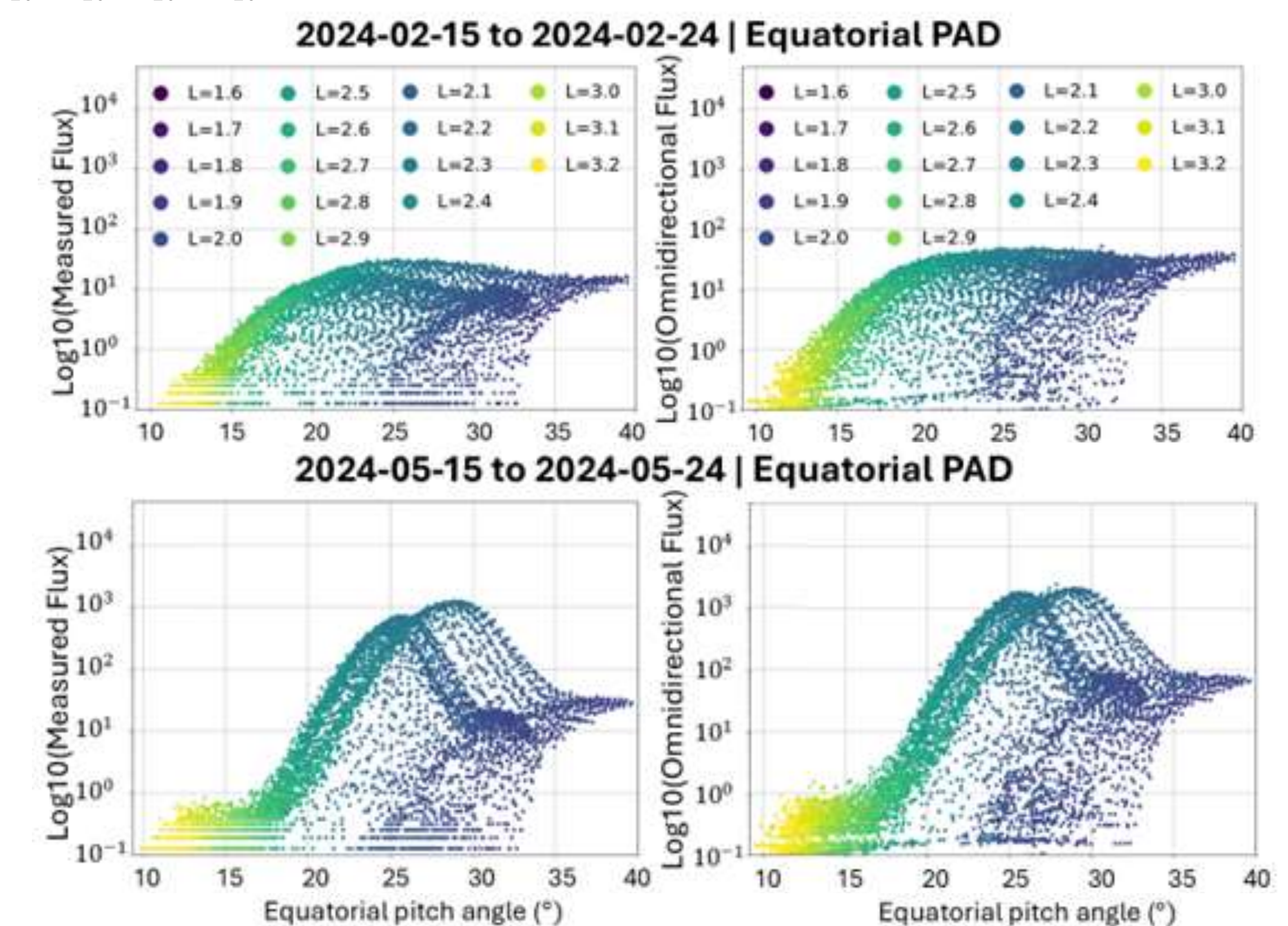


Figure 13: Equatorial Pitch-Angle Distributions of (left panel) measured and (right panel) retrieved omnidirectional flux for the time periods (top) 15-24 February 2024, (bottom) 15-24 March 2024. Colour indicates L-shell.

- Two humps arising from the northbound/southbound passes become **much more aligned in magnitude** after retrieval, as expected from the method.
- Quiet-time shape: smooth rise from low flux near 10° to a broad peak. Storm-time: around an order-of-magnitude enhancement with two peaks near 25° and 30° .

Conclusions:

- The anisotropy index n is fairly stable over time in the heart of the inner belt, but shows strong dependence on geomagnetic activity near the outer edge.
- Raw directional fluxes underestimate omnidirectional flux by **at least half an order of magnitude** - confirming the need for high-fidelity response modelling for accurate LEO radiation interpretation.
- Future work:** extend to additional NGRM energy channels; apply the same framework to Sentinel-6B and MetOp-SG (also carrying NGRM).

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